

ALLERGY

Skin damage signals mediate allergic sensitization to spatially unlinked antigen

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Our current understanding of immunity to pathogens suggests that anatomic coupling of antigens with danger signals is a required feature for the formation of immune memory. However, in the context of pathogen-independent inflammation, the stringency of this anatomical coupling is unclear. Here, we demonstrate that multiple modes of skin injury were sufficient to induce a humoral response to antigens introduced in the gut. Skin damage induced a narrow subset of endocrine cytokines that were necessary and sufficient for the priming of antigens introduced at various distal tissues. Thus, in addition to “local priming” of antigen entering through damaged skin, there also exists another paradigm of “remote priming” where anatomical coupling is not essential because of the dissemination of damage-associated intermediaries. Our findings have implications for understanding the fundamental mechanisms of the formation of humoral memory with wide implications for diseases such as food allergy and in vaccinology.

INTRODUCTION

Despite strong baseline tolerogenic mechanisms, immunity can be inappropriately licensed, resulting in breaches of tolerance that manifest as allergic or chronic inflammatory diseases. The pathogenesis of this inappropriate licensing is not well understood. Experimental mouse models often use adjuvants, which are compounds that induce danger-coded costimulation signals, to induce resistance mechanisms that generate immunity rather than tolerance (1). Adjuvants often act through multiple and/or unknown mechanisms, making it difficult to tease apart the pathways important for licensing an immune response (2). Classic examples of allergic adjuvants are aluminum salts and cholera toxin, whose mechanisms of action are still unclear despite decades of intense investigation (3, 4). Additionally, experimental adjuvants are always mixed with the antigen of interest in an effort to mimic the spatial proximity of these two immunologic signals during a microbial exposure. However, whether this physical proximity during delivery is a requirement for immunologic sensitization has not been explored and becomes important when identifying physiologic drivers of disease. Knowledge of these licensing parameters could provide insights into the pathogenesis of autoimmune and allergic diseases in humans.

We were particularly interested in whether inflammation at a distal tissue, such as the skin, could result in sensitization to novel oral antigens rather than tolerance, which is the default outcome in the gut (5). Various inflammatory skin conditions are associated with non-skin-specific diseases, including food allergy, inflammatory

bowel disease, and arthritis (6–9). In the case of food allergy, it has been proposed that acutely or chronically damaged skin provides a “natural” adjuvant effect for allergens on the skin to generate allergen-specific immunoglobulin E (IgE) and subsequent anaphylaxis upon allergen ingestion (10). Although proximity between antigen and skin damage is sufficient to induce allergic sensitization, it remains unclear whether spatial linkage is necessary, especially considering that food antigens primarily enter via the gastrointestinal route (11–18).

Skin inflammation is often thought to be either self-contained or a consequence of underlying systemic inflammation. However, it is increasingly recognized that skin inflammation can induce systemic inflammation resulting in immunopathology at distal sites, as exemplified by the associations between psoriasis and metabolic or cardiovascular disease (19, 20). Moreover, acute injury of the skin induces the release of skin epithelium-derived signals into the blood, which can reprogram the intestinal or lung immune environment to enable anaphylaxis or asthma, respectively (15, 21–24). Thus, in addition to initiating local antigen-specific responses in the skin (“local priming”), cutaneous inflammation can also coordinate systemic responses that drive antigen-independent immunopathology in other tissues.

However, prior studies have only shown that these endocrine signals are sufficient to modify an ongoing immune or tissue response. They have not demonstrated whether these endocrine signals are capable of initiating a de novo, nontolerogenic, adaptive memory response. We hypothesized that skin injury could serve as a natural adjuvant to prevent the establishment of tolerance and rather induce sensitization to novel antigens encountered at distal portals of entry. In this study, we found that the systemic effects of skin inflammation are sufficient to initiate humoral responses to spatially uncoupled antigens, a concept that we define as “remote priming.”

RESULTS

Local skin inflammation can allow for remote priming of humoral immunity to bystander antigens in the gut

The traditional model of epicutaneous sensitization is a form of local priming wherein antigen is introduced to the site of tape-stripped, damaged skin (14, 15, 18). Given the systemic inflammatory effects

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of skin damage, we hypothesized that damage to the cutaneous barrier could alter the trajectory of novel antigens introduced via the noninflamed gastrointestinal tract from a tolerogenic to an immunogenic response. To test whether remote priming could occur, we first altered the traditional tape-stripping epicutaneous allergic sensitization model by introducing the model food protein ovalbumin (OVA) via the gastrointestinal route by oral gavage rather than topically via damaged skin after inducing acute skin injury by tape stripping (Fig. 1A). Unexpectedly, the spatial unlinking of adjuvant (skin damage) from antigen was sufficient to induce the production of allergy-associated OVA-specific IgG1 and total IgE isotypes but not those associated with an antimicrobial response (IgM, IgG2b/c, and IgG3) (Fig. 1B and fig. S1A). In accordance with these elevated allergic antibodies, mice exhibited an anaphylactic response upon systemic OVA challenge as indicated by a brisk and robust temperature drop (Fig. 1C). Thus, skin damage-induced remote priming generates isotype-switched antibodies of sufficient quantity and quality to be functional in vivo.

To test whether this phenomenon was unique to tape stripping or a more generalizable feature of skin injury, we modeled two other distinct forms of acute injury: ulcerating plaques, induced by intradermal acetone injection, and penetrative wounds from a punch biopsy. Using the same sensitization scheme as before (Fig. 1A), where the antigen was introduced orally but not in the skin, both of these acute injury models similarly induced OVA-specific IgG1 and IgE (Fig. 1B and fig. S1, B and C) as well as an anaphylactic response to systemic OVA challenge (Fig. 1, D and E).

Given that these acute skin injuries were sufficient to induce allergic sensitization, we next wondered whether remote priming could occur with chronic skin damage. We used the calcipotriol (MC903) model of atopic dermatitis, which has previously been shown to allow for local priming in the skin (17), and tested its capacity for remote priming to OVA in the gut. This model of atopic dermatitis was also sufficient to sensitize mice as evidenced by increased OVA-specific IgG1, OVA-specific IgE, total IgE (Fig. 1B and fig. S1D), and anaphylaxis upon systemic OVA challenge (Fig. 1F).

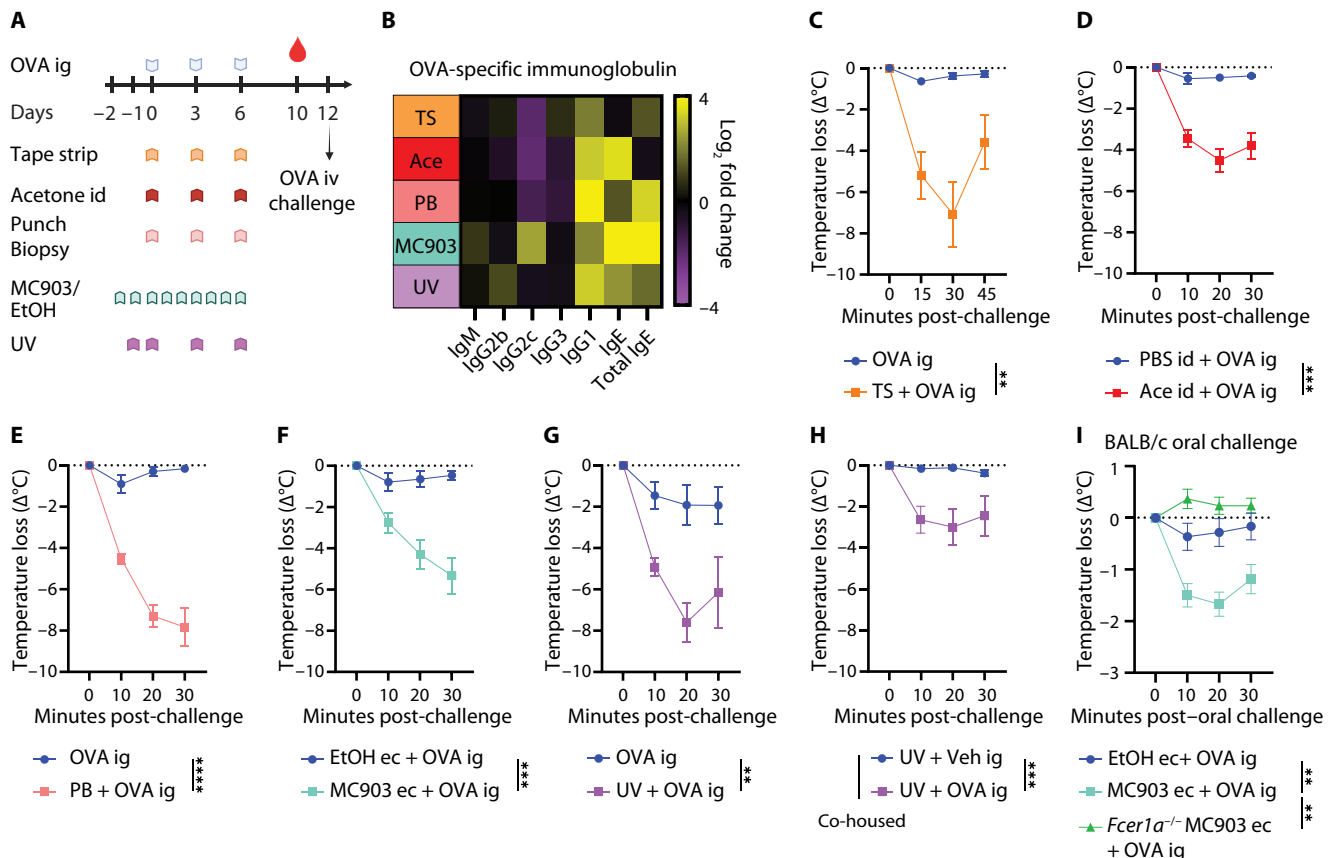


Fig. 1. Skin inflammation can prime humoral responses to bystander antigens in the gut. (A) Schematic representation of skin injury-induced oral sensitization. Mice were subjected to one of various acute injury modalities [tape stripping (TS), intradermal acetone injection (Ace), punch biopsy (PB), severe UV exposure, or chronic atopic dermatitis (MC903)]. OVA was administered intragastrically (ig) immediately after the acute injury events or topical application of chronic inflammatory agent on days 0, 3, and 6. Plasma was collected on day 10, and mice were challenged with OVA intravenously (iv) on day 12. (B) Heatmap of OVA-specific humoral responses as log₂ fold change over respective controls [depilated mice for TS, PB, and UV; intradermal (id) PBS for ACE; and epicutaneous (ec) ethanol (EtOH) for MC903]. (C to H) Temperature loss induced by intravenous challenge for each skin damage modality. (H) Mice were exposed to UV, given OVA or vehicle (Veh) intragastrically, and kept cohoused on wire-bottom cages during sensitization. (I) Temperature loss induced by oral challenge in BALB/c mice on day 12. ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001, using two-way analysis of variance (ANOVA) with the [(C) to (H)] Šidák or (I) Tukey multiple comparisons correction. Data are presented as means ± SEM (*n* = 4 to 10 mice per group) and are representative of at least two independent experiments, with the exception of (F), which are pooled from two experiments.

Last, we tested whether forms of skin damage previously associated with nonallergic types of inflammation could also induce sensitization. Exposure to ultraviolet (UV) radiation, for example, is associated with increases in type 1 inflammatory signals [interleukin-1 β (IL-1 β) and interferon- γ (IFN- γ)] and plasmacytoid dendritic cells (DCs) (25–27). Nonetheless, high-dose UV exposure combined with oral OVA exposure (Fig. 1A) could also induce anaphylaxis upon systemic OVA challenge (Fig. 1G) and an increase in OVA-specific IgG1 and OVA-specific IgE (Fig. 1B and fig. S1E).

UV exposure damages a large area of skin, which may enable local priming through contaminating environmental OVA in the feces, bedding, or during self or group grooming. To confirm that OVA exposure occurs via the gastrointestinal tract and not through unintended skin exposure, we cohoused UV-exposed and OVA-gavaged mice with UV-exposed and vehicle-gavaged mice in wire-bottom cages. Anaphylaxis and the generation of OVA-specific antibodies were wholly dependent on the introduction of OVA by gavage given that cohoused mice that did not receive oral OVA were not sensitized by residual OVA in the environment (Fig. 1H and fig. S1F). Similar results were observed for mice whose skin was damaged with intradermal acetone or punch-biopsy wounds (fig. S2, A and B). Thus, various forms of skin damage can induce remote priming of the gut rather than inadvertent local priming of the skin.

The commensal microbiota is another possible contributor to inflammation originating from the skin. We therefore assessed this directly by comparing acetone-induced remote priming in specific pathogen-free (SPF) mice and germ-free (GF) mice devoid of any fungal or bacterial commensals. Like SPF mice, GF mice underwent systemic anaphylaxis and produced OVA-specific IgG1 and IgE after this treatment (fig. S2, C and D). Thus, microbiota are not a general requirement for remote priming. Rather, the licensing signals for remote priming are intrinsic to skin damage itself.

Because of the spatially unlinked nature of this sensitization paradigm, we were curious whether the convergence of the adjuvant signals and protein signals occurred in a canonical fashion. Using lymphotoxin- α -deficient mice (*Lta*^{-/-}), in which secondary lymphoid organs are severely atrophic, we observed a complete loss of remote priming potential in response to tape stripping (fig. S2E). Thus, skin damage-intrinsic signals induce a targeted adaptive response in secondary lymphoid organs and are not simply an unorganized bystander effect of skin inflammation.

Priming that followed all forms of skin damage largely induced antibody isotypes associated with allergy. To further establish the allergic nature of skin damage-induced remote priming, we evaluated whether anaphylaxis was dependent on the classic antibody mediator IgE. Tape stripping-induced remote priming required IgE for anaphylaxis (fig. S2E). The type 2-dominant nature of the effector response was further explored in mice deficient in IL-4, which is critical for IgG1- and IgE-isotype switching. As anticipated, IL-4 deficiency abrogated OVA-specific IgG1 and IgE generation and anaphylactic responses (fig. S2, F to H). Although systemic challenges had already established the *in vivo* functionality of the generated OVA-specific antibodies, we also wanted to determine the potential relevance of this sensitization paradigm to food allergy. Given the associations between atopic dermatitis and food allergy, we subjected BALB/c mice, which are amenable to oral challenge, to the MC903-induced atopic dermatitis model. We remotely primed BALB/c mice sufficient or deficient in the high-affinity IgE receptor Fc ϵ RI α and challenged them by oral gavage on days 12 and 28, after

the skin inflammation had resolved. Both immediately after sensitization (Fig. 1I) and after recovery (fig. S2I), oral challenge induced a temperature drop in an Fc ϵ RI α -dependent manner, suggesting that remote priming can allow for canonical oral anaphylaxis independent of ongoing inflammation. Thus, many forms of skin inflammation are capable of licensing the intestinal immune environment from a tolerizing milieu into one that promotes allergic sensitization.

Circulating skin injury-derived cytokines are sufficient to mediate remote priming

We next investigated the mechanism by which anatomically distal signals (i.e., a damage signal at the skin and foreign antigen at the gut) would converge to generate immunological memory. We reasoned that skin damage might generate endocrine signals that could contextualize antigens concurrently sampled at distal sites. We therefore profiled the systemic cytokine response by measuring plasma cytokines after injury on days 0 or 3 (Fig. 2A and fig. S3, A to D). Tape stripping induced the release of not only IL-33, as anticipated (15), but also IL-1 β , IL-6, IL-18, and IL-21. Intradermal acetone injection resulted in a moderate elevation of IL-9 and IL-33 and a modest increase in IL-4, monocyte chemoattractant protein-1 [also known as CCL2 (chemokine ligand 2)], and tumor necrosis factor (TNF). Punch biopsy induced a modest elevation of IL-6 and IL-33. Atopic dermatitis from MC903 treatment yielded abundant levels of thymic stromal lymphopoietin (TSLP), as was expected (28), as well as IL-1 β , IL-3, IL-4, IL-7, and IL-33. By contrast, UV exposure was associated with elevated levels of cytokines characteristic of type 1 (IL-1 β and IFN- γ), type 2 (IL-4, IL-13, IL-25, IL-33, and TSLP), and type 3 (granulocyte colony-stimulating factor, IL-6, and IL-17A). Thus, skin damage induces “codes” of endocrine signals that communicate the nature of damage systemically.

Although different types of skin injury induced varying combinations of cytokines in the blood, they were all capable of inducing allergic sensitization to orally introduced antigens. We consequently wondered whether any of these circulating cytokines were sufficient to initiate humoral immunity. We therefore developed a reductionist remote-priming protocol in which single recombinant cytokines resolved from our plasma analysis were injected intravenously, whereas OVA was injected into the footpads of mice for ease of screening (Fig. 2B). Mice sensitized with different intravenous cytokines were then assessed for their production of OVA-specific antibodies and their body temperature loss in response to systemic OVA challenge (Fig. 2C and fig. S3, E and F). Most of the cytokines tested were unable to induce the generation of OVA-specific antibodies or drive anaphylaxis. Although IL-9-, IFN- γ -, and IL-4c-treated mice produced low levels of OVA-specific IgG1 and TNF- α -treated mice generated OVA-specific IgM and IgG2b, none of these animals underwent anaphylaxis upon systemic challenge (fig. S3, E and F). IL-33, TSLP, IL-1 β , and IL-18 were the only cytokines that induced both OVA-specific IgG1 and IgE and resulted in an anaphylactic response after systemic challenge. Additionally, IL-1 β and IL-18 were able to robustly induce the production of other IgG isotypes. This highlights three patterns of cytokine activity: (i) adjuvanticity without skewing activity, such as IL-1 β and IL-18, which make all antibody isotypes; (ii) adjuvanticity with skewing activity, such as IL-33, TSLP, and TNF- α , which direct formation of a specific class of antibody isotypes; and (iii) skewing activity lacking intrinsic adjuvanticity, such as IL-4 and IL-13, which cannot induce antibodies but are essential in defining which isotypes are made. Each skin

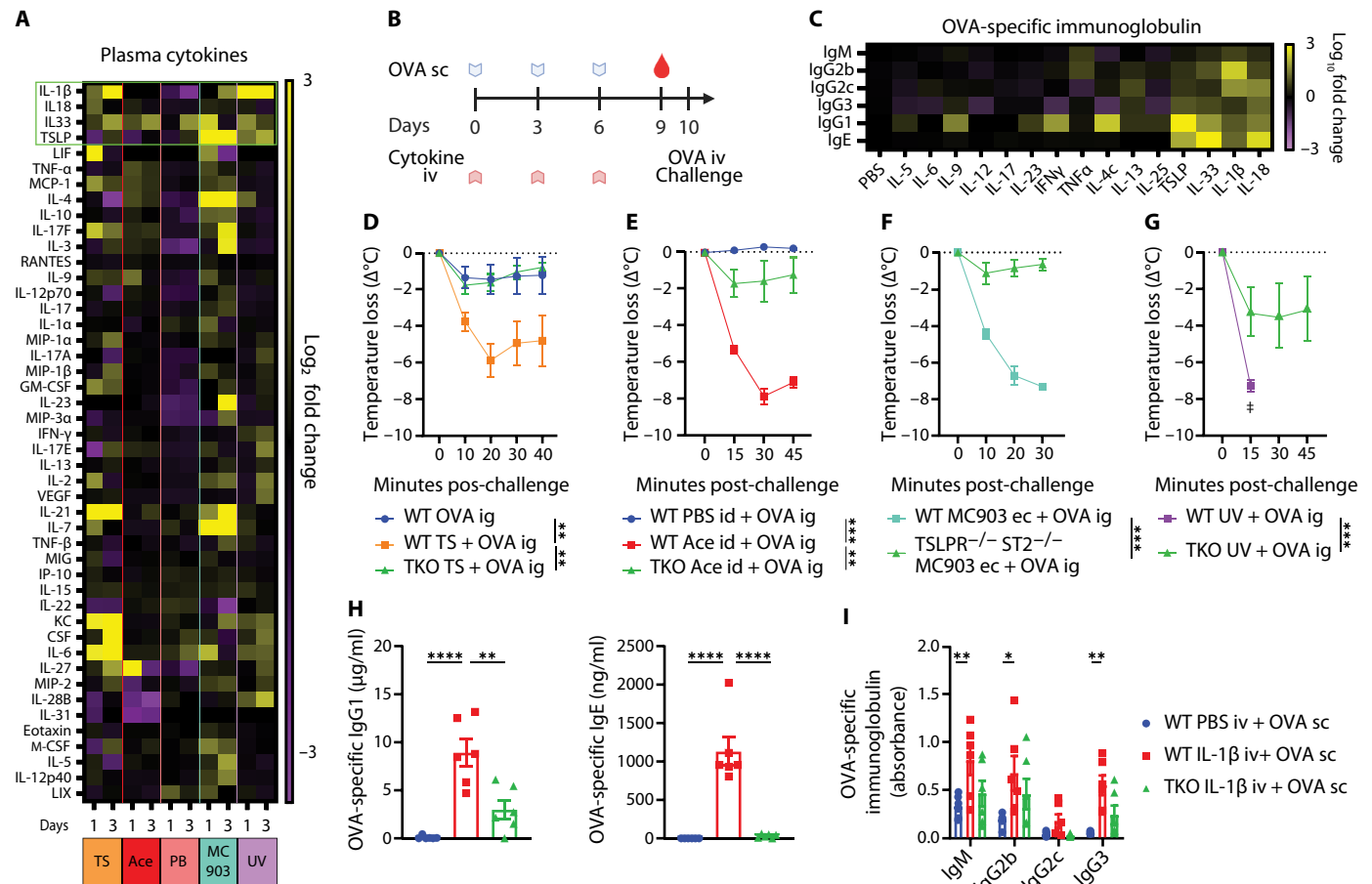


Fig. 2. Circulating skin injury-derived cytokines mediate remote priming in a coded fashion. (A) Multiplex cytokine analysis from plasma 1 hour after skin injury or topical treatment on days 0 and 3. IL-33, TSLP, IL-1 β , and IL-18 were measured by ELISA. LIF, leukemia inhibitory factor; RANTES, regulated on activation, normal T cell expressed and secreted; MIP, macrophage inflammatory protein; GM-CSF, granulocyte-macrophage colony-stimulating factor; VEGF, vascular endothelial growth factor; MIG, monokine induced by interferon-gamma [also known as CXCL9 (C-X-C motif chemokine ligand 9)]; IP-10, interferon-gamma inducible protein 10 kDa (also known as CXCL10); KC, keratinocyte chemoattractant (also known as CXCL1); CSF, colony-stimulating factor; M-CSF, macrophage colony-stimulating factor; and LIX, lipopolysaccharide-induced CXC chemokine (also known as CXCL5). Values reflect the log₂ fold change over the average of depilated mice or respective vehicle controls [PBS intradermally or ACE and EtOH epicutaneously for MC903]. (B) Schematic representation of systemic cytokine-induced distal sensitization. Mice were sensitized with 0.5 to 1 μ g of recombinant cytokine iv and 10 μ g of OVA subcutaneously (sc) on days 0, 3, and 6; bled on day 9; and challenged with OVA intravenously on day 10. (C) Relative OVA-specific immunoglobulin concentrations reflected as log₁₀ fold change over the average of the PBS controls. (D to G) Temperature loss after 100- μ g OVA intravenous challenge on day 10 after the indicated skin injury-induced sensitization in wild-type (WT) and ST2-TSLPR-IL-25 triple-knockout (YRS TKO) mice. # denotes death of group during challenge. (H and I) OVA-specific immunoglobulins after sensitization with 1 μ g of IL-1 β iv and 10 μ g of OVA sc in WT and TKO mice. * P < 0.05, ** P < 0.01, *** P < 0.001, and **** P < 0.0001, using [(D), (E), and (I)] two-way ANOVA with the Tukey multiple comparisons correction, [(F) and (G)] unpaired t test at first time point, or (H) one-way ANOVA with the Tukey multiple comparisons correction. Data are presented as means $\pm$ SEM (n = 2 to 13 mice per group). With the exception of the screens in (A) and (B), which were performed once, data are representative of at least two independent experiments.

damage modality induces one or more of these IgE-generating cytokines, highlighting a shared feature across these modes of injury that could result in remote priming (Fig. 2A and fig. S3, A to D). Each skin damage modality resulted in a different pattern of IgG1 and IgE induction, suggesting that these cytokines are not acting in isolation but are influenced by the remaining cytokine milieu that is unique to each modality (Figs. 1B and 2A). MC903, for example, has a concomitant high presence of IL-4, which may bolster the adjuvant-potent TSLP to induce the highest levels of OVA-specific IgE among all of the skin damage-induced remote priming models.

Because IL-33 and TSLP were each sufficient to reproduce the same pattern of elevated OVA-specific IgG1 and IgE but equivalent

IgG2b/c and IgG3 that we observed with our skin injury sensitizations, we thought that they might serve as the primary initiating pathways in these models. Moreover, IL-33 and TSLP, along with IL-25, form a family of cytokines colloquially referred to as allergic “alarmins” that are released in response to epithelial damage and play a crucial role in the initiation of local allergic inflammation (29). Because alarmins often act in a redundant fashion, we decided to use mice that were double-knockouts (DKOs) of the IL-33 receptor (ST2/IL-1RL1) and the TSLP receptor (TSLPR) or ST2-TSLPR-IL-25 triple-knockout (TKO) mice to test their roles in skin injury sensitization. Loss of the alarmin pathways largely abrogated the anaphylactic response to systemic OVA challenge and associated

OVA-specific IgG1 or IgE for tape stripping-, intradermal acetone-, MC903-, and UV-induced remote priming (Fig. 2, D to G, and fig. S4, A to D). MC903- and acetone-induced remote priming was independent of IL-25, whereas compensation was observed between all three alarmins in the tape-stripping and UV modalities (fig. S4, E to G).

Alarmins were necessary for the production of allergic isotypes despite the simultaneous presence of IL-1 β and/or IL-18 in the plasma of tape-stripped or UV-injured wild-type mice (Fig. 2A). Moreover, mice deficient in caspases 1/4, which serve as the canonical activators of IL-1 β and IL-18, fully retained the ability to be primed (fig. S4, F and G). This led us to hypothesize that IL-1 β and IL-18 are dependent on the alarmins for remote priming rather than being a completely independent pathway. After remote priming by either IL-1 β or IL-18 and intraplantar OVA, alarmin TKO mice exhibited substantially abrogated induction of OVA-specific IgG1 and IgE (Fig. 2I and fig. S5, A and B). Using mice with different combinations of alarmin deficiencies, we observed that most of the IL-1 β -induced remote priming to IgG1 and IgE was dependent on TSLPR signaling (fig. S5, C and D). These data suggest that alarmins serve a critical role in the generation of antigen-specific IgG1 and IgE from composite systemic cytokine combinations induced by skin injury or from simplified remote priming by IL-1 β or IL-18. Thus, endocrine alarmin signals generated by skin damage are necessary and sufficient to mediate remote priming.

Circulating IL-33 is sufficient to induce high-affinity IgE to novel antigens introduced at any barrier site

The observation of remote priming immediately led us to ask how indiscriminate priming would be averted, especially because immune memory responses are highly durable and could last a lifetime. We expected that signal-to-noise discrimination would be essential to reduce the potential for indiscriminate remote priming of coincident antigens in the setting of skin damage. Because of the critical role of IL-33 during skin damage and its higher potency to induce OVA-specific IgE rather than the other cytokines (fig. S3F), we chose to further investigate the signal-to-noise features of remote priming using our reductionist model with recombinant IL-33 as an adjuvant. We hypothesized that this discrimination would depend on the dose and timing of antigen and cytokine adjuvant as well as any previously established tolerance. By sensitizing with intravenous IL-33 and intraplantar OVA (Fig. 2B), we observed that the production of OVA-specific IgE is dependent on the dose of both IL-33 as well as that of OVA (Fig. 3, A to D). IL-33-induced remote priming of the footpad was ST2 dependent, produced a durable OVA-specific IgG1 and IgE response, and retained anaphylactic potential a month after discontinuing sensitization when there was no longer any IL-33 activity that might directly promote anaphylaxis (fig. S6, A and B). Thus, both the dose of antigen and adjuvant require exceeding certain thresholds to result in a durable memory response.

Having established quantitative thresholds of adjuvant and antigen, we next wanted to define the temporal constraints of remote sensitization. We consequently tested the effect of increasing time lapses between intravenous IL-33 and intraplantar OVA injections (Fig. 3, E and F). Maximal OVA-specific IgG1 and IgE were observed when the administration was separated by 2 hours. Even after a 6-hour delay between injections, when IL-33 was no longer detectable in plasma (fig. S6C), OVA-specific IgG1 and IgE were still unexpectedly induced. To assess the affinity of the IgE that was

induced by IL-33, we sensitized T follicular helper (T_{FH}) cell-sufficient (*Bcl6^{fl/fl}*) and T_{FH} cell-deficient mice (*Cd4-cre⁺Bcl6^{fl/fl}*) with intravenous IL-33 and intraplantar 4-hydroxy-3-nitrophenyl acetyl (NP)-haptenated OVA. NP-specific IgE bound to both the densely (NP27) and sparsely (NP2) coated antigens in a T_{FH} cell-dependent manner (Fig. 3G), suggesting that IL-33-induced IgE is of high affinity as a consequence of a mature adaptive response.

Because intraplantar injection of OVA does not represent a normal route of entry for food antigens, we performed a similar sensitization scheme with intravenous IL-33 but introduced OVA via the gut or lungs. High levels of OVA-specific IgG1 and IgE were both induced regardless of the portal of entry of OVA (Fig. 3H and fig. S6D). We then evaluated how many doses were required and observed that both routes produced OVA-specific IgG1 after one immunization dose but required a minimum of two sensitizations in order for animals to class-switch to OVA-specific IgE (Fig. 3, I and J, and fig. S6E). Remote priming by IL-33 was not restricted to the purified OVA protein, given that blended peanuts could also be used as the oral antigen to generate peanut-specific IgG1 and IgE (fig. S6F). This phenomenon of remote priming was also not constrained to mice on a B6 background, as BALB/c mice were also able to generate OVA-specific IgG1 and IgE (fig. S6, G and H) and anaphylax to oral OVA (Fig. 3K). This is consistent with previous reports that IL-33 can prime the tissue environment (accumulation of intestinal mast cells, eosinophils, and ILC2s) and influence the activation threshold of these cells to enable oral anaphylaxis (15, 21, 30). Last, the remote priming of intestinal OVA occurred only when it was introduced as a novel antigen, because the generation of OVA-specific IgE and anaphylaxis was abrogated if the mouse had consumed OVA in their drinking water before sensitization (Fig. 3L and fig. S6I). Thus, although circulating IL-33 has the capacity to license an allergic rather than tolerogenic response to antigen in the gut, it is unable to break previously established tolerance.

These data indicate that use of intravenous cytokine adjuvants provides a highly tunable and versatile model for allergic sensitization. Moreover, we established a model of allergic sensitization with a molecularly defined model antigen and a clinically relevant, recombinant protein adjuvant, which is in contrast with the complex or nonphysiologic chemical compositions now used as adjuvants in most published food allergy models.

IL-33 remodels the immune landscape to promote IgE production in part via ILC2s

To understand the mechanism by which IL-33 mediates remote priming, we first analyzed the immune composition of secondary lymph organs (SLOs) during sensitization. Intravenous injection of IL-33 caused an exposure-dependent increase in the ST2⁺ populations in SLOs (fig. S7A). Basophils, eosinophils, and group 2 innate lymphoid cells (ILC2s) express both ST2 and IL-4 (31–36). During the course of IL-33-induced remote priming of intraplantar OVA, all three populations were increased in the spleen, draining lymph node, and nondraining lymph nodes. After the third immunization, basophils contracted, eosinophils stabilized, and ILC2s expanded even further (Fig. 4A and fig. S7B). Because of the expansion of these IL-4-competent ST2⁺ cells, we wondered whether there was a relationship between IL-33 and IL-4. As anticipated, the production of OVA-specific IgG1 and IgE during IL-33 remote priming was completely dependent on the presence of IL-4 (fig. S7C). However, the absence of IL-4 caused a shift to IgM and other IgG isotypes (fig.

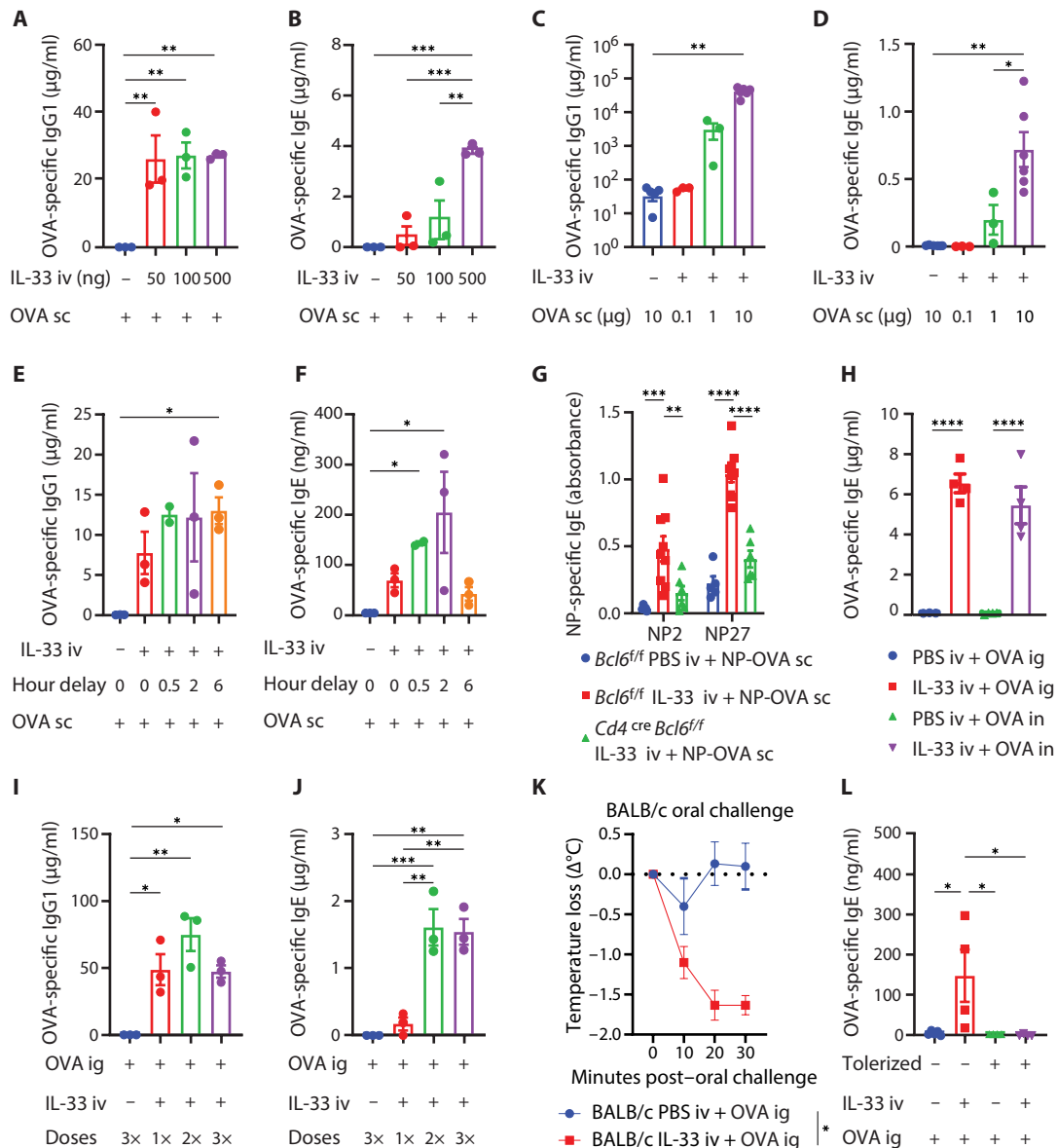


Fig. 3. Circulating IL-33 is sufficient to induce high-affinity IgE to novel antigens at any barrier site. (A, C, and E) OVA-specific IgG1 and (B, D, and F) OVA-specific IgE after distal sensitization with [(A) and (B)] increasing doses of recombinant IL-33 intravenously and 10 μ g of OVA sc in the footpad, [(C) and (D)] 100 ng of IL-33 iv with increasing doses of OVA via subcutaneous footpad injection, or [(E) and (F)] 100 ng of IL-33 iv and rested for the indicated times before 10 μ g of OVA via sc footpad injection. (G) BSA-NP-2–specific (high-affinity) and BSA-NP-27–specific (low-affinity) IgE from T_{FH} cell-sufficient (*Bcl6*^{fl/fl}) or T_{FH} cell-deficient mice (*Cd4*^{cre}*Bcl6*^{fl/fl}) that were immunized with 100 ng of IL-33 iv and 10 μ g of NP-20–coated OVA sc. Plasma was assayed at 1:10 dilutions for all samples. (H) OVA-specific IgE of mice distally sensitized with 500 ng of IL-33 iv and either 50 mg of OVA ig or 100 μ g of OVA intranasally (in), as indicated. (I) OVA-specific IgG1 and (J) OVA-specific IgE of mice immunized with 100 ng of IL-33 iv and 50 mg of OVA ig on day 0 (1x); days 0 and 3 (2x); or days 0, 3, and 6 (3x), as indicated. (K) Temperature loss induced by the fourth oral challenge of BALB/c mice remotely primed with 100 ng of IL-33 iv and 50 mg of OVA ig. (L) OVA-specific IgE of naïve or OVA–“tolerized” mice immunized with 100 ng of IL-33 iv and 50 mg of OVA ig. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001, using one-way ANOVA with the [(A), (B), (H), (I), (J), and (L)] Tukey or (E) Dunnett multiple comparisons correction, [(C), (D), and (F)] Kruskal-Wallis test with the Dunn multiple comparisons correction, or two-way ANOVA with the (G) Tukey or (K) Šidák multiple comparisons correction. Data are presented as means $\pm$ SEM (*n* = 3 to 9 mice per group) and are representative of at least two independent experiments.

S7D), suggesting that the ability to initiate humoral immunity is intrinsic to IL-33 itself, whereas the ability to class-switch to IgE is not. We therefore hypothesized that IL-33 acts directly on ST2-expressing immune cells to make the IL-4 that is then required for IgE skewing.

In addition to the cells we observed expanding in SLOs after IL-33 exposure, many other immune cells, such as myeloid and T

cells, also express ST2. Accordingly, we screened the ability of various global and conditional knockouts to undergo IL-33–induced remote priming of intragastric OVA. Germline deficiencies in eosinophils (Δ dblGATA), IL-1 β /18 activation (*Casp 1/4*^{−/−}), or diphtheria toxin–mediated depletion of mast cells and basophils (RMB) had no effect on IL-33 remote priming (fig. S7, E to G). Moreover,

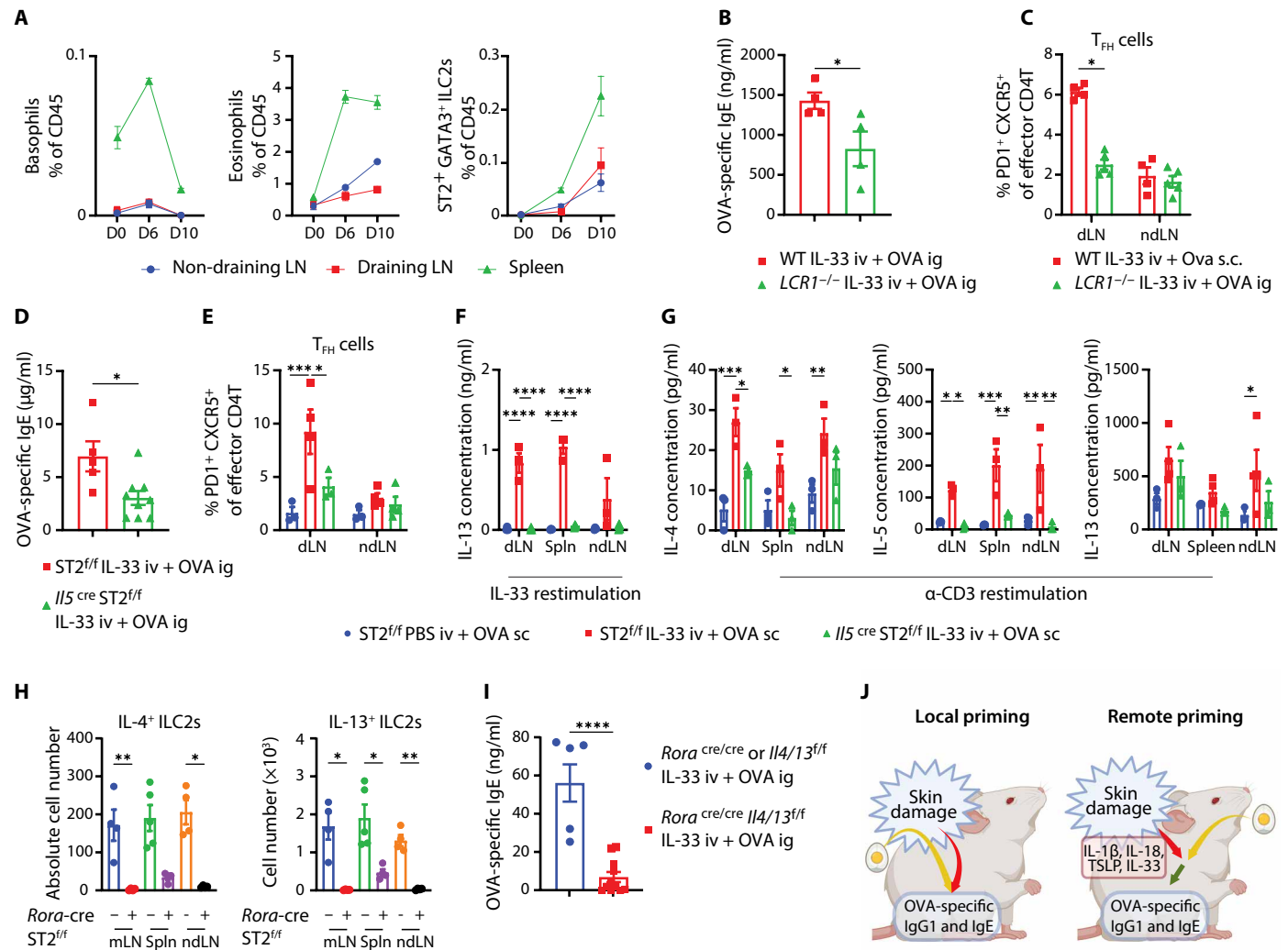


Fig. 4. IL-33 remodels the immune landscape to promote allergic sensitization, in part, via ILC2s. (A) Quantification of CD49b⁺FcεR1α⁺ basophils, CD11b⁺Siglec-F⁺ eosinophils, and Lin⁺KLRG1⁺Thy1⁺ST2⁺GATA3⁺ ILC2s as percent of CD45⁺ cells in indicated organs and days after sensitization with 100 ng of IL-33 iv and 10 μg of OVA sc. *n* = 2 to 4 mice per group. D, days. (B, D, and I) OVA-specific IgE from the indicated mice on day 9 after sensitization with 100 ng of IL-33 iv and 50 mg of OVA ig. (C and E) T_{FH} cell percentages of CD62L⁺CD44⁺ effector CD4 T cells in indicated organs at day 6 after sensitization with 100 ng of IL-33 iv and 10 μg of OVA sc. (F and G) IL-4, IL-5, and/or IL-13 measured in the supernatant of cells after ex vivo stimulation on day 10 with (F) IL-33 or (G) anti-CD3. (H) Quantification of IL-4⁺ or IL-13⁺-expressing Lin⁺KLRG1⁺ST2⁺ ILC2s in indicated organs on day 10 after sensitization with 100 ng of IL-33 iv and 10 μg of OVA sc. (J) Local priming involves the introduction of antigen at the same place, such as via damaged or inflamed skin. Remote priming occurs when damage or inflamed skin induces the release of damage-associated cytokines with adjuvant potential, namely IL-1β, IL-18, TSLP, and IL-33, which can act systemically to initiate a humoral response to antigens that enter at any site, such as the gastrointestinal tract. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001, using [(B), (D), and (I)] unpaired *t* test, two-way ANOVA with the (C) Šidák correction or [(E) to (G)] Tukey multiple comparisons correction, (H) the Kruskal-Wallis *H* test with the Dunn multiple comparisons correction (for IL-4⁺), or (H) the Brown-Forsythe and Welch test (for IL-13⁺). Data are presented as means ± SEM (*n* = 2 to 11 mice per group) and are representative of at least two independent experiments with the exception of (I), which are pooled from two experiments. LN, lymph node; mLN, mesenteric lymph node; dLN, draining lymph node; ndLN, non-draining lymph node; Spln, spleen.

conditional ablation of ST2 in T cells (*Cd4-Cre*), DCs (*CD11c-Cre* and *Zbtb46-Cre*), or myeloid cells (*LysM-Cre*) did not alter the production of OVA-specific IgG1 or IgE (fig. S8, A to D). The efficiency of *Zbtb46-Cre* deletion of ST2 on DCs was high, and, although *CD11c-Cre*-mediated deletion only appeared to be moderate, it was sufficient to prevent the generation of PDL2⁺ or CD301b⁺ allergy-associated DCs after intraplantar immunization of IL-33 and OVA (fig. S8, E and F). Thus, eosinophils, mast cells, basophils, T cells, and a large subset of myeloid cells including relevant DC

populations important for allergic sensitization were all dispensable for sensing IL-33 during remote priming.

Given that the ST2⁺ ILC2 population increased dramatically, we next assessed their role in IL-33 remote priming in the gut. Using *LCR1*-knockout mice (37), which are selectively deficient in ILC2s, we observed a significant decrease in OVA-specific and total IgE production, although the IgG1 production was unaffected (Fig. 4B and fig. S9, A and B). This loss of OVA-specific IgE in the *LCR1*^{-/-} mice was associated with a decreased induction of T_{FH} cells at day 6

of remote priming of the footpad sensitization in the draining, but not nondraining, lymph nodes (Fig. 4C).

We subsequently used a conditional deletion approach to target ILC2s using *Il5*-Cre ST2^{fl/fl} mice, whose ILC2s are severely impaired in their ability to expand upon IL-33 administration (fig. S9C). As anticipated, *Il5*-Cre ST2^{fl/fl} mice exhibited a partial reduction in total and OVA-specific IgE as well as OVA-specific IgG1 upon IL-33 remote priming of the gut (Fig. 4D and fig. S9, D and E). *Il5*-Cre ST2^{fl/fl} mice also showed reductions in OVA-specific IgG1 and IgE during IL-33 remote priming in the footpad (fig. S9F). In addition to a global ILC2 deficiency, *Il5*-Cre ST2^{fl/fl} mice had a substantial deficiency in T_{FH} cell generation during remote priming of the footpad (Fig. 4E and fig. S9G). Conditional ablation of ST2 using the *Rora*-Cre, which has primarily been shown to target ILC2s and CD4 T cells, led to an almost complete loss of OVA-specific IgG1 and IgE after IL-33 remote priming of the gut, similar to the use of *Vav1*-Cre, which targets all leukocytes (figs. S8D and S9, H and I). This loss was again accompanied by an impairment in T_{FH} cell generation during IL-33 remote priming of the footpad (fig. S9J). Because ablation of ST2 in T cells using the *Cd4*-Cre does not result in an impairment of OVA-specific IgE or T_{FH} cell formation (figs. S8A and S9K) and deletion of ILC2s was sufficient to abrogate OVA-specific IgE, we expected that any off-target effect of ST2 deletion in T cells by the *Il5*- or *Rora*-Cre was not responsible for the diminished ability of IL-33 to drive OVA-specific IgE. *Il5*-Cre ST2^{fl/fl} mice were also unable to produce IL-13 or IL-5 in response to ex vivo stimulation of lymph nodes with IL-33 after remote priming of the footpad (Fig. 4F and fig. S9L). Ex vivo restimulation of these lymph nodes with anti-CD3 similarly led to a diminished capacity to generate IL-4 and IL-5, which suggested a defect in T helper 2 (T_{H2}) cell induction (Fig. 4G).

Lastly, previous work has demonstrated a role for ILC2-derived IL-4/13 in modifying allergic inflammation (15, 38); therefore, we sought to determine its relevance in our model. After IL-33 remote priming of the footpad in *Il4*-mRNA reporter mice, there was a notable percentage of ILC2s transcribing *Il4* (fig. S9M). We further validated this by intracellular cytokine staining in *Rora*-Cre ST2^{fl/fl} mice, wherein IL-4-producing ILC2s accumulated in an ST2-dependent fashion (Fig. 4H and fig. S9N). The production of IL-4 and IL-13 in *Rora*⁺ cells was required, given that *Rora*-Cre *Il4/Il13*^{fl/fl} mice had a complete loss of OVA-specific IgE (Fig. 4I). Thus, the deletion of ST2 in ILC2s decreased the overall tone of IgE-skewing cytokines within the lymph node in both a cell-intrinsic and a cell-extrinsic manner, the latter via the impaired generation of both T_{FH} and T_{H2} cells. Thus, endocrine IL-33 directly causes the expansion of ILC2s in all lymphoid compartments, and these ILC2s are nonredundantly required for the switching to IgE, likely because of their broad effects on T_{FH} and T_{H2} cell generation and their production of IL-4/13.

DISCUSSION

Here, we report that various types of acute and chronic skin inflammation can lead to the formation of immune memory to antigens coincidentally introduced at distal portals of entry, including the highly tolerogenic environment of the gut, a phenomenon that we call remote priming (Fig. 4J). The universality of the responses to generate allergic isotypes reflects the fact that skin damage can generally serve as a natural type 2 adjuvant both locally and systemically.

We explored one mechanism of how remote priming occurs on the basis of the release of soluble signals from the damaged skin into circulation. Although different types of skin damage induced different systemic inflammatory profiles, they all contained TSLP, IL-33, IL-1 β , or IL-18, which act as damage-associated molecular patterns (DAMPs), indicating tissue or cellular damage (39). These comprise a unique subset of cytokines that, when present in the blood, are each sufficient on their own to induce an IgE response to spatially unlinked antigens, consistent with their reported roles in sensitizing spatially linked antigens (40). The endocrine cytokine signaling pathway that we have described here likely works in concert with other mechanisms that we have not addressed, such as the mobilization of cells out of the damaged skin or the transmission of neuronal signals, to fully communicate signals from damaged skin to the gut (41–44).

We provide mechanistic dissection of IL-33 remote priming and identify ILC2 sensing via ST2 as being required in a nonredundant manner for the maximal generation of IgE. The contribution of ILC2s may occur by two parallel mechanisms. The first is upstream of effector T cell generation, such as by promoting a DC migration or activation state (45–48). Additionally, ILC2-derived IL-4/13 promotes mast cell expansion in the intestine and may act in lymphoid organs to supplement the predominantly T_{FH} cell-derived IL-4/13 to further enhance B cell class switching to IgE (49, 50). Moreover, the incomplete loss of IgE indicates that there still remains a yet to be identified *Rora*⁺ cell that is critical for initiating the humoral response to the IL-33 signal that is not affected by the DC, myeloid, or T cell targeting strategies that we used. Because of the off-target effects of the *Rora*-Cre, we also cannot ascertain the individual contributions of IL-4 or IL-13 derived from ILC2s, T cells, or the original IL-33-sensing *Rora*⁺ cell. The use of an intravenous cytokine adjuvant has provided several advantages. First, by using a single signal that can initiate a sensitizing humoral response, rather than a complex mixture from skin damage or adjuvant, we could more directly assess the signaling cascades responsible for sensitization. Second, this approach allows for the introduction of antigens in any location, imparting great versatility to studying immunizations or disease models at different sites. This paradigm also has implications for vaccine development, which could be complementary or synergistic with vaccine strategies such as “prime and spike” to boost mucosal-resident defenses (51). Last, by having full control of the signals going into the model organism, we can begin to investigate intercytokine dynamics and generate cytokine networks that encode specific immune responses (52) and resolve the inflammatory networks in human disease that support allergic sensitization and those that do not (e.g., atopic dermatitis versus psoriasis), even if they share a common signal like IL-33 (6, 53–55).

Attempts to delineate a set of pattern-associated molecular patterns (PAMPs) for allergic inflammation has yielded few candidates, none of which are universal (56). By contrast, the role of DAMPs in the form of alarmins has been repeatedly demonstrated for local priming (30, 40). Our data support a layered paradigm where, in the absence of PAMPs, DAMPs (57) provide a second layer of recognition initiating an immune response. Although the specificity of PAMPs has been trained through the course of evolution, DAMP-induced initiation requires training within each individual. As we have demonstrated with IL-33, the training by remote priming is reminiscent of classical conditioning of neurologic memory. There is an unconditioned stimulus (skin damage) that is

tied to a bystander conditioned object (distal antigen) to provoke a conditioned response (allergic memory). The linking of the unrelated stimulus and object is a balance of signal to noise (doses of antigen and damage signals), requires repetition, and is weakened by increased time and previously learned responses (tolerance). Associative training provides an opportunity for anticipatory responses in instances where the insult itself does not introduce specific antigens, such as a chemical irritant or burn, but is consistently correlated with environmental antigens that can be targeted (58–60). Remote priming allows for the environmental antigen-sensing capacity to be broadened to all mucosal barriers, which increases the pool from which to identify the antigen with the best signal-to-noise ratio that correlates to the damage ensued. Future work will be necessary to fully decode the dynamics and nature of combinatorial cytokine signaling in shaping remotely primed humoral responses and for the role this may play in health and disease.

MATERIALS AND METHODS

Study design

The objective of this study was to investigate how skin inflammation could cause sensitization to food antigens entering into the gut. We used several models of acute skin injury or inflammation in mice to assess their ability to generate oral antigen-specific allergic antibodies and subsequently their anaphylactic response to antigen challenge. Upon observing a release of cytokines in the serum of damaged mice, we developed a model to screen for individual adjuvant potential of various cytokines when introduced intravenously coincident with oral antigen. Sample sizes were determined by preliminary experiments. Outliers were defined with Grubbs or ROUT (robust regression and outlier removal) tests or removed from statistical analysis at certain time points if they reached the end point early. All mouse experiments were designed with random assignment of mice without investigator blinding, with at least three biological replicates per experimental group. The number of mice and statistical methods used are described in the figure legends.

Animals

All animal experiments were performed in accordance with institutional regulations after protocol review and approval by Yale University's Institutional Animal Care and Use Committee. Animals were used between the ages of 8 and 10 weeks and maintained in an SPF facility. The following strains were obtained from the Jackson Laboratory: C57BL/6J (no. 000664), BALB/cJ (no. 000651), *Zbtb46*-cre/zDC-cre [B6.Cg-*Zbtb46*^{tm3.1(cre)Mnz}/J] (no. 028538), CD11c-cre [C57BL/6J-Tg(*Itgax*-cre,EGFP) 4097Ach/J] (no. 007567), LysM-Cre [B6.129P2-*Lyz2*^{tm1(cre)Jfo}/J] (no. 004781), *Vav1*-cre [B6.Cg-*Commd10*^{Tg(Vav1-cre)A2Kio}/J] (no. 008610), Δ dblGATA [B6.129S1(C)-*Gata1*^{tm6Sho}/LytzJ] (no. 033551), *Casp1*^{1/4}-^{fl/fl} (B6N.129S2-*Casp1*^{tm1Flv}/J) (no. 016621), *Lta*^{-/-} (B6;129S2-*Lta*^{tm1Dch}/J) (no. 002257), *Il4*^{-/-} (C57BL/6-*Il4*^{tm1Nnt}/J) (no. 002518), and *Il5*-cre [B6(C)-*Il5*^{tm1.1(cre)Lky}/J] (no. 030926). *Cd4*-cre *Bcl6*^{fl/fl} mice and *Il4*-GFP/*Il21*-Katushka mice were obtained from the Craft lab (Yale University) (61). *Ighe*^{-/-} (B6 background) and *Fcer1a*^{-/-} (BALB/c background) animals were obtained from R. Medzhitov. *Il1rl1*^{fl/fl} and *Cd4*-cre⁺*Il1rl1*^{fl/fl} mice were obtained from A. Y. Rudensky (Rockefeller) (62). R. M. Locksley provided YRS (*Arg1*^{YFP}*Il5*^{tdTomato}*Il13*^{hCD4}) ILC2-reporter mice on an ST2-TSLPR-IL-25 TKO (*Il1rl1*^{-/-}*Crlf2*^{-/-}*Il25*^{-/-}) (TKO) C57BL/6 background (63). *Il1rl1*^{-/-} C57BL/6 mice were from A. N. J. McKenzie. W. J. Leonard provided the *Crlf2*^{-/-} mice

on a C57BL/6 background (64). The ST2-TSLPR DKO mice were generated by crossing the *Il1rl1*^{-/-} mice with the *Crlf2*^{-/-} mice. GF mice on a C57BL/6 background were obtained from the Palm lab and housed in bioexclusion cages (Tecniplast ISO72P). *LCR1*^{-/-} mice were from J. Henao-Mejia and M. F. Michieletto from the University of Pennsylvania (37). RMB (B6.Ms4a2tm1Mal) mice were provided by P. Launay (Université Paris Diderot) (65). *Rora*-Cre and *Il4/Il13*^{fl/fl} mice on a BALB/c background were obtained from R. Geha (15). *Rora*-Cre mice were backcrossed onto B6-*Il1rl1*^{fl/fl} mice for four generations, with Cre⁻ control mice derived from the fourth backcross used as controls.

Mast cell and basophil depletion

RMB mice were administered 1 μ g of diphtheria toxin (Sigma-Aldrich, D0564) in 100 μ l of phosphate-buffered saline (PBS) intraperitoneally on days -6, -4, and -2. They were then administered 500 ng in 100 μ l of PBS intraperitoneally on days 2, 5, and 8 to thoroughly deplete mast cells and basophils before and throughout sensitization.

Skin damage sensitization

Regardless of the type of acute injury or chronic inflammation, mice were gavaged with 50 mg of grade V OVA (Sigma-Aldrich, A5503) in 200 μ l of 5% sodium carbonate monohydrate (Sigma-Aldrich, 230952) immediately after injury or topical application on days 0, 3, and 6 (scheme in Fig. 1A). Mice were bled on day 10 and allowed to rest for 2 days before being challenged intraorbitally with 100 μ g of endotoxin-free OVA (InvivoGen, vac-pova-100) in 100 μ l of PBS on day 12. Oral challenge in BALB/c mice was performed on days 12 and 28 by gavage of 50 mg of OVA in 200 μ l of 5% sodium carbonate monohydrate. Temperature was recorded using a rectal probe (Thermalert TH-5) every 10 to 15 min as indicated.

Mechanical skin injury

The backs of 8- to 12-week-old male mice were dorsally shaved at least 1 day before the initiation of tape stripping-induced mechanical injury. Tape stripping consisted of applying and ripping off tape (Scotch Super Hold 700S10) on a 2 cm-by-2 cm area of the shaved back until erythema was visible (~20 to 25 times). Tape stripping was performed on days 0, 3, and 6 on different areas of the back each day. Female mice were not as reproducibly consistent with tape stripping and were, therefore, excluded from this specific form of sensitization only.

Intradermal acetone-induced ulcerating plaques

The backs of 8- to 12-week-old male and female mice were shaved and depilated with Nair (746122459660) 1 day before beginning sensitization. Different areas of back skin were injected intradermally with 100 μ l of PBS or acetone (JT Baker 9006-05) on days 0, 3, and 6.

Punch biopsy wounds

The backs of 8- to 12-week-old male and female mice were shaved and depilated with Nair 1 day before beginning sensitization. Punch biopsies were performed on days 0, 3, and 6 on different parts of the back under isoflurane anesthesia. While anesthetized, the skin was sterilized with 70% alcohol and then pinched at the rostral and caudal ends to generate a fold of skin along the mouse's spine. The folded skin was then flipped and pressed against a styrofoam block, and a 4-mm Punch Biopsy (Acuderm Inc., P425) was pushed through both sides of folded skin, generating two bilateral punches on each day. Iodine solution (McKesson, 911740) was applied to wound sites to prevent infection.

Calcipotriol-induced atopic dermatitis

The backs of 8- to 12-week old male and female mice were shaved and depilated with Nair on day -3. Mice were treated epicutaneously once daily with ethanol or 50 μ l of 100 μ M calcipotriol/MC903 (MedChemExpress, HY-10001) dissolved in ethanol from day -2 until day 6. To prevent the mice from licking the treatments off the skin, the solution was allowed to evaporate before returning the mice to the cage.

High-dose UV exposure

The backs of 8- to 12-week-old male and female mice were shaved and depilated with Nair on day -2. On days -1, 0, 3, and 6, the mice were exposed to broadband UV radiation (450 mJ/cm²). Mice in a wire mesh-covered cage were placed into a UV box custom-made by the Yale engineering department with TL 20 W/12 RS Phillips broadband UV-B light bulbs emitting radiation at 290 to 315 nm. The duration of exposure to achieve the 450 mJ/cm² dose was calibrated using a UV intensity meter (G&R Labs Model 100).

Recombinant cytokine remote priming

Mice were immunized three times with carrier-free recombinant murine cytokines in 100 μ l of PBS retro-orbitally and OVA (route as indicated in figure legends) on days 0, 3, and 6. They were then bled on day 9 and challenged intravenously with 2 μ g of endotoxin-free OVA. Temperature was recorded using a rectal probe as indicated. The cytokines and doses used can be found in table S1. Unless otherwise indicated, subcutaneous footpad injections were administered as 10 μ g of endotoxin-free OVA (InvivoGen, vac-pova-100) in 50 μ l of PBS. For NP-specific experiments, mice were injected in the footpad with 10 μ g of NP-20 OVA (Biosearch, N-5051-100) in 50 μ l of PBS. Intranasal administration was performed under ketamine (Ketaset) and xylazine (Anased) anesthesia using 100 μ g of endotoxin-free OVA (InvivoGen, vac-pova-100) in 50 μ l of PBS. Intragastric administration was with 50 mg of grade V OVA (Sigma-Aldrich, A5503) or 5 mg of freshly ground blanched raw peanut (Aiva Products) in 200 μ l of 5% sodium carbonate monohydrate (Sigma-Aldrich, 230952). Peanuts were pulsed in a Shardor Coffee grinder no. CG7155 into a coarse meal before being further ground into a paste with the vehicle and back of syringe. Oral challenge in BALB/c mice was performed on days 10, 12, 14, and 16 with 50 mg of grade V OVA in 200 μ l of 5% sodium carbonate monohydrate. OVA tolerization was achieved by providing 1% grade V OVA in the drinking water (ad libitum) during 4 to 8 weeks of age and then removed 1 week before sensitization.

Plasma antibodies

Antibodies were quantified by enzyme-linked immunosorbent assay (ELISA). IgM, IgG1, IgG2b, IgG2c, and IgG3 were measured by coating NUNC plates (Thermo Fisher Scientific, 442404) with OVA (20 μ g/ml; InvivoGen, vac-pova-100) in carbonate coating buffer (Sigma-Aldrich, C3041) overnight at 4°C. Total and OVA-specific IgE plates were coated with anti-mouse IgE (BD Pharmingen, 553413; clone R35-72). NP-specific IgG1 and IgE plates were coated with NP2- or NP27-bovine serum albumin (BSA; 1 μ g/ml; Biosearch N-5050XL-10, N-5050H-10). Peanut-specific IgG1 and IgE plates were coated in crude peanut extract (20 μ g/ml; Greer Laboratories) (49). Plates were washed in 0.05% Tween 20 (American Bio) and blocked in 1% BSA (Sigma-Aldrich, A9647) for a minimum of 1 hour at room temperature. Plasma samples were diluted in 1% BSA and incubated overnight at 4°C. The following standards were used:

total IgE-purified mouse IgE (BD Pharmingen, 557080), OVA-specific IgE (Bio-Rad, MCA2259), and mouse OVA-specific IgG1 (Chondrex, 3008). Detection was performed for 1 hour at room temperature with the following detection antibodies: anti-mouse IgM (Abcam, 97230), anti-mouse IgG1-biotin (BD Pharmingen, 553441), anti-mouse IgG2b-horseradish peroxidase (HRP; Abcam, 97250), IgG2c-HRP (Abcam, 97255), IgG3-HRP (Abcam 97260), anti-mouse IgE-biotin (BD Pharmingen, 553419; clone R35-118), and biotinylated OVA (Nanocs, OVA-1-BN-1). For NP- and peanut-specific IgE, plasma for all treatments and genotypes was diluted 1:10 in blocking buffer, and detection was done by anti-mouse IgE-HRP (SouthernBiotech, 1130-05). Streptavidin HRP was incubated for 30 min before washing and developing with TMB (BD OptEIA, 555214) and stopped with 1 mol/liter HCl. Concentrations for ELISA reagents can be found in table S2. Plates were read at 450 nm on a spectrophotometer (Biotek Synergy HT) with (Gen5) software.

Plasma cytokines from skin damage

Plasma was collected 1 hour after indicated injuries or topical treatment on days 0 and 3 of treatment. Plasma was sent to Eve Technologies to analyze for circulating cytokines (MD32 and MDT_H17-12). To generate a heatmap, the injury values were divided by the average values for the respective controls for each time point. The fold change was then log₂ transformed, and each cell on the heatmap represented the average of the three experimental samples for each type of damage at a given time point. Individual cytokine ELISAs for IL-1 β (R&D Systems, MLB00C), IL-18 (Invitrogen, BMS618-3), IL-33 (R&D Systems, M3300), and TSLP (R&D Systems, MTL00) were performed according to the manufacturer's instructions.

Flow cytometry

Ipsilateral inguinal and popliteal lymph nodes were pooled to form draining lymph nodes. Contralateral axillary and brachial lymph nodes were pooled to form nondraining lymph nodes. Spleens were treated with red blood cell lysis buffer (BioLegend, 420302). For DC-specific isolations, lymph nodes were collected at 18 hours and digested in collagenase IV (0.2 mg/ml; Worthington) and deoxyribonuclease I (0.05 mg/ml; Sigma-Aldrich) at 37°C for 1 hour. Single-cell suspensions were then washed once in PBS with 2% fetal bovine serum (FBS) and incubated with blocking solution and viability dye. The cells were washed twice and stained for 30 min on ice. The anti-murine cell-surface antibodies used can be found in table S2. Lineage (lin) was defined by CD4, CD3, CD8, B220, CD19, TCR γ d, Ly6C, Siglec-F, IgE, NK1.1, NKp46, CD11b, CD11c, and CD64. ILC2s were defined as CD45⁺lin⁻Thy1⁺Sca-1⁺CD127⁺KLRG1⁺GATA3⁺ cells. For intracellular GATA3 staining, the cells were fixed and permeabilized using the FOXP3-fixation kit (eBioscience, 00-5523-00) according to the instructions and stained for 4 hours at 4°C.

For intracellular cytokine staining, cells were stimulated with cell activation cocktail (BioLegend, 423302) for 2 hours at 37°C. BD GolgiPlug was then added (BD Biosciences, 555029), and cells were incubated for an additional 3 hours at 37°C. Cells were surface-stained as described above and then fixed with BD Cytofix/Cytoperm (BD Biosciences, 554722) for 15 min at room temperature. Intracellular staining was performed in permeabilization buffer (Invitrogen, 00-8333-56) overnight at 4°C. Cells were analyzed using a FACSymphony A5 flow cytometer (BD Biosciences). The data were analyzed with FlowJo software v10.8.1.

Restimulation

Single-cell suspensions of lymph nodes and spleens were collected on day 10 of IL-33 remote priming. Five hundred thousand cells were plated in complete RPMI medium and 10% FBS, with or without anti-CD3 (1 µg/ml) or IL-33 (100 ng/ml). Supernatants were collected after 48 hours and analyzed by ELISA for IL-4, IL-5, and IL-13 (Invitrogen, 88-7044-88, 88-7054-88, and 88-7137-88).

Statistics

All statistical analyses were performed in GraphPad Prism 7. Statistical information is included in the figure legends. For anaphylaxis experiments, tests were performed on time points where all mice were alive, and significance is displayed for the most significant time point. Graphics were generated using BioRender.

Supplementary Materials

The PDF file includes:

Figs. S1 to S9
Tables S1 and S2
Legend for data file S1

Other Supplementary Material for this manuscript includes the following:

Data file S1
MDAR Reproducibility Checklist

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Acknowledgments: We would like to thank A.Y. Rudensky, J.M. Leyva-Castillo, J.E. Craft, R. Medzhitov, R.M. Locksley, A.N.J. McKenzie, W.J. Leonard, P. Launay, and R.S. Geha for providing the mice that made this project possible. **Funding:** This work was supported by the Francis Trudeau Trainee Fellowship (D.A.W.); Food Allergy Science Initiative Award FASI-000611 (A.W.); National Institutes of Health grants R01 AI162645 and R01 AR080104 (A.W.) and K08 AI171182-0 (A.E.); Pew Biomedical Scholars Award (A.W.); Smith Family Foundation Odyssey Award (A.W.); Charles Hood Family Foundation Award (A.W.); Colton Center for Autoimmunity at Yale Award (A.W. and A.E.); Chan Zuckerberg Institute Award (A.W.); Knights of Columbus Award (A.W.); Robert Leet and Clara Guthrie Patterson Trust Award (A.E.); Yale Center for Clinical Investigation/Yale Physician Scientist Development Award (A.E.); Dermatology Foundation Career Development Award (A.E.); and Doris Duke Fund to Retain Clinical Scientists (A.E.). **Author contributions:** Conceptualization: D.A.W., A.E., and A.W. Methodology: D.A.W., A.E., and A.W. Investigation: D.A.W., A.E., I.B.-S., A.L.M., C.Z., Y.M., K.Z., K.I.-W., and A.W. Funding acquisition: A.W. Project administration: C.Z. Resources: R.M., P.L., M.F.M., J.H.-M., N.W.P., and J.C. Supervision: A.W. and A.E. Writing—original draft: D.A.W. Writing—review and editing: D.A.W., A.E., A.W., K.I.-W., K.Z., and I.B.-S. **Competing interests:** A.W. consulted for NGM Biopharmaceuticals, Seranova Biopharmaceuticals, The Column Group, and the Knights of Columbus during the period in which this work was produced. All other authors declare that they have no competing interests. **Data and materials availability:** All data needed to evaluate the conclusions in the paper are present in the paper or the Supplementary Materials. Tabulated underlying data for Figs. 1 to 4 and figs. S2 to S4 and S6 are archived in data file S1. Requests for RMB (B6.Ms4a2^{tm1Mad}) mice can be fulfilled by contacting P.L. (Université Paris Diderot, pierre.launay@inserm.fr).

Submitted 20 November 2023

Resubmitted 10 June 2024

Accepted 13 March 2025

Published 4 April 2025

10.1126/sciimmunol.adn0688